

Lesson 01 · Quantities, percentages and algebra

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

Without using a calculator, evaluate $10 \times (-2)$.

☐ A -20

☐ B 20

☐ C -12

☐ D 12

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Q2 · Rank 1 / Foundation

Without using a calculator, evaluate $7 - (-4)$.

☐ A 3

☐ B -11

☐ C 11

☐ D -3

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Q3 · Rank 1 / Foundation

Without using a calculator, find the value of $2x - y$ when $x = 7$ and $y = 4$.

☐ A 6

☐ B 18

☐ C 3

☐ D 10

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Q4 · Rank 2 / Core

Simplify the following algebraic expression by collecting like terms: $6x + 5y - 2x - 12y$.

☐ A $4x - 7y$

☐ B $4x + 17y$

☐ C $8x - 7y$

☐ D $-3xy$

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Q5 · Rank 2 / Core

Reduce the numerical fraction $\frac{18}{30}$ to its lowest terms.

☐ A $\frac{9}{10}$

☐ B $\frac{3}{5}$

☐ C $\frac{5}{3}$

☐ D $\frac{2}{5}$

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Q6 · Rank 2 / Core

Calculate 5% of 24.

☐ A 12

☐ B 0.12

☐ C 1.2

☐ D 120

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Q7 · Rank 3 / Application

In 2011 in the United States, 35 out of every 100 adults owned a smartphone. By 2013 this figure increased to 56 out of every 100. By what factor did smartphone ownership increase during this period? Give your answer as a mixed fraction in its lowest terms.

☐ A $\frac{5}{8}$ ☐ B $1\frac{3}{5}$ ☐ C $\frac{3}{5}$ ☐ D $2\frac{1}{10}$

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Q8 · Rank 3 / Application

A firm has 132 female and 88 male employees. What percentage of staff are female?

☐ A 40%☐ B 66%☐ C 150%☐ D 60%

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Q9 · Rank 3 / Application

A student discount card reduces a bill in a restaurant from USD 124 to USD 80.60. Work out the percentage discount.

☐ A 35%☐ B 65%☐ C 53.85%☐ D 43.40%

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Q10 · Rank 4 / Challenge · OPTIONAL

A firm has 132 female and 88 male employees. During the next year 8 additional female staff are employed. If the percentage of female staff is now 56%, how many additional male staff were recruited during the year?

☐ A 22☐ B 30☐ C 8☐ D 12

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